

Modelling of Stochastic Processes

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1 Equations Of Motion

From Apostolos derivations we have:

$$\ddot{\phi} + 3H\dot{\phi} - \frac{\kappa V_0}{bM_{\text{Pl}}} \exp\left[\frac{\phi}{bM_{\text{Pl}}} - \kappa e^{\frac{\phi}{bM_{\text{Pl}}}}\right] + g^2\langle\chi^2\rangle(\phi - \phi_{\text{SBP}}) = 0 \quad (1)$$

$$\ddot{\tilde{\chi}_{\mathbf{k}}} + 3H\dot{\tilde{\chi}_{\mathbf{k}}} + \left[\frac{k^2}{a^2} + g^2(\phi - \phi_{\text{SBP}})^2 - \lambda f^2 + 3\lambda\langle\chi^2\rangle + \langle\dot{\theta}^2\rangle + \frac{1}{a^2}\langle(\nabla\theta)^2\rangle\right]\tilde{\chi}_{\mathbf{k}} = 0 \quad (2)$$

$$\langle\chi^2\rangle\ddot{\tilde{\theta}_{\mathbf{k}}} + (2\langle\chi\dot{\chi}\rangle + 3H\langle\chi^2\rangle)\dot{\tilde{\theta}_{\mathbf{k}}} + \frac{k^2}{a^2}\langle\chi^2\rangle\tilde{\theta}_{\mathbf{k}} = 0 \quad (3)$$

where the averages are

$$\langle\chi^2\rangle = \int_{\mathbb{R}^3} \frac{d^3p}{(2\pi)^3} [|\tilde{\chi}_{\mathbf{p}}(t)|^2 + 2\text{PI-corrections}]$$

$$\langle\dot{\theta}^2\rangle = \int_{\mathbb{R}^3} \frac{d^3p}{(2\pi)^3} [|\dot{\tilde{\theta}}_{\mathbf{p}}(t)|^2 + 2\text{PI-corrections}]$$

I want to rewrite these equations in a "vector" form for numerical integration, such that for instance

$$x = \tilde{\chi}_{\mathbf{k}} \quad \text{and} \quad v = \dot{\tilde{\chi}_{\mathbf{k}}}$$

so that

$$X = \begin{bmatrix} x \\ v \end{bmatrix} = \begin{bmatrix} \tilde{\chi}_{\mathbf{k}} \\ \dot{\tilde{\chi}_{\mathbf{k}}} \end{bmatrix}$$

Then

$$\dot{v} + 3Hv + \left[\frac{k^2}{a^2} + g^2(\phi - \phi_{\text{SBP}})^2 - \lambda f^2 + 3\lambda\langle\chi^2\rangle + \langle\dot{\theta}^2\rangle + \frac{1}{a^2}\langle(\nabla\theta)^2\rangle\right]x = 0$$

To get a better understanding of these equations I will first compact some terms. Changing (1) into:

$$\ddot{\phi} + 3H\dot{\phi} - V(\phi, \langle\chi\rangle) = 0$$

with

$$V(\phi, \langle\chi^2\rangle) = \frac{\kappa V_0}{bM_{\text{Pl}}} \exp\left[\frac{\phi}{bM_{\text{Pl}}} - \kappa e^{\frac{\phi}{bM_{\text{Pl}}}}\right] + g^2\langle\chi^2\rangle(\phi - \phi_{\text{SBP}})$$

Then (2)

$$\ddot{\tilde{\chi}_{\mathbf{k}}} + 3H\dot{\tilde{\chi}_{\mathbf{k}}} + \omega\tilde{\chi}_{\mathbf{k}} = 0$$

$$\omega = \omega(\phi, \langle\chi^2\rangle, \langle\dot{\theta}\rangle, \langle(\nabla\theta)^2\rangle) = \frac{k^2}{a^2} + g^2(\phi - \phi_{\text{SBP}})^2 - \lambda f^2 + 3\lambda\langle\chi^2\rangle + \langle\dot{\theta}^2\rangle + \frac{1}{a^2}\langle(\nabla\theta)^2\rangle$$

and for (3) I am pretty sure I can just divide by $\langle\chi^2\rangle$

$$\ddot{\tilde{\theta}_{\mathbf{k}}} + \left(2\frac{\langle\chi\dot{\chi}\rangle}{\langle\chi^2\rangle} + 3H\right)\dot{\tilde{\theta}_{\mathbf{k}}} + \frac{k^2}{a^2}\tilde{\theta}_{\mathbf{k}} = 0$$

To make this managable computable, I think I need to first compute the averages of all.

2 Simpler equations

It makes sense to first start simulations with simpler equations where we take $\langle \dot{\theta} \rangle = \langle (\nabla \theta)^2 \rangle = 0$ and forget about the third equation entirely.

$$\ddot{\phi} + 3H\dot{\phi} - \frac{\kappa V_0}{bM_{\text{Pl}}} \exp \left[\frac{\phi}{bM_{\text{Pl}}} - \kappa e^{\frac{\phi}{bM_{\text{Pl}}}} \right] + g^2 \langle \chi^2 \rangle (\phi - \phi_{\text{SBP}}) = 0 \quad (4)$$

$$\ddot{\tilde{\chi}_{\mathbf{k}}} + 3H\dot{\tilde{\chi}_{\mathbf{k}}} + \left[\frac{k^2}{a^2} + g^2(\phi - \phi_{\text{SBP}})^2 - \lambda f^2 + 3\lambda \langle \chi^2 \rangle \right] \tilde{\chi}_{\mathbf{k}} = 0 \quad (5)$$

where we can go a step further and first consider the homogeneous equation

$$\ddot{\phi} + 3H\dot{\phi} - \frac{\kappa V_0}{bM_{\text{Pl}}} \exp \left[\frac{\phi}{bM_{\text{Pl}}} - \kappa e^{\frac{\phi}{bM_{\text{Pl}}}} \right] = 0 \quad (6)$$

To compute the averages we must discretise the integral over $\mathbf{k}$ modes, and first we will not consider any 2PI corrections

$$\langle \chi^2 \rangle = \int_{\mathbb{R}^3} \frac{d^3 p}{(2\pi)^3} |\tilde{\chi}_{\mathbf{p}}(t)|^2 \approx \sum_{\mathbf{p}} \omega_{\mathbf{p}} |\tilde{\chi}_{\mathbf{p}}(t)|^2$$

where $\omega_{\mathbf{p}}$ being the weights. There are several choices of $\omega_{\mathbf{p}}$, but since $\langle \chi^2 \rangle$ is not isometric we use

$$\omega_{\mathbf{k}} = \frac{\delta k_x \delta k_y \delta k_z}{(2\pi)^3}$$

which is determined at the start of the simulation when we choose the change of k , assuming that the range is divided in equal steps that is $\delta k_x = \delta k_y = \delta k_z = \delta k \equiv \text{const.}$

$$\omega_{\mathbf{k}} = \left(\frac{\delta k}{2\pi} \right)^3 = \omega$$

yielding the simplified expression

$$\langle \chi^2 \rangle \approx \omega \sum_{\mathbf{p}} |\tilde{\chi}_{\mathbf{p}}(t)|^2$$

Another important thing to note is that the hubble parameter H also changes, since it depends on the energy density ρ of the fields

$$\begin{aligned} \rho_{\phi}(t) &= \frac{1}{2} \dot{\phi}^2 + V(\phi) \\ \rho_{\chi}(t) &= \int \frac{d^3 k}{(2\pi)^3} \frac{1}{2} (|\dot{\tilde{\chi}_{\mathbf{k}}}|^2 + \omega_k) \end{aligned}$$

Then $\rho_{\text{tot}} = \rho_{\phi} + \rho_{\chi}$, however at first we can neglect the ρ_{χ} term and the hubble parameter thus becomes

$$H \approx \sqrt{\frac{1}{3M_{\text{Pl}}^2} \left(\frac{1}{2} \dot{\phi}^2 + V(\phi) \right)} = \sqrt{\frac{1}{3M_{\text{Pl}}^2} \left(\frac{1}{2} \dot{\phi}^2 + V_0 \exp \left[-\kappa e^{\frac{\phi}{bM_{\text{Pl}}}} \right] \right)}$$

Note that we are using the potential $V(\phi) = V_0 \exp \left[-\kappa e^{\frac{\phi}{bM_{\text{Pl}}}} \right]$.

Next, I need to determine initial conditions. It makes sense to integrate from the symmetry breaking point ϕ_{SBP} . During kination (point where energy density is dominated by kinetic energy) the ϕ field equation simplifies immensely:

$$\ddot{\phi} + 3H\dot{\phi} = 0$$

with a simplified Hubble parameter $H \approx \frac{|\dot{\phi}|}{\sqrt{6}M_{\text{pl}}}$. We can then solve these equations. Note that we will take $t_i = t_{\text{SBP}} = 0$ and $v = \dot{\phi}_{\text{SBP}}$

$$\begin{aligned}\phi &= \sqrt{\frac{2}{3}} \text{sign}(v) \ln \left(1 + \sqrt{\frac{3}{2}} \frac{|v|t}{M_{\text{pl}}} \right) M_{\text{pl}} \\ \dot{\phi} &= v \exp \left(-\sqrt{\frac{3}{2}} \frac{\phi}{M_{\text{pl}} \text{sign}(v)} \right)\end{aligned}$$

That means that initial conditions $\phi_0 = \phi(t_i) = 0$ and $\dot{\phi}_0 = \dot{\phi}(0) = v$, where I think I can choose v - being a free parameter. We can similarly consider modes for χ field. For large $|\phi|$ they satisfy

$$\ddot{\chi}_k + 3H\dot{\chi}_k + \omega_k^2 \chi_k = 0$$

where $\omega^2 = k^2 + g^2 \phi^2 - \lambda f^2$. The initial conditions are given by the Bunch-Davies vacuum state

$$\chi_{\phi, \text{vac}} = \frac{a^{-1}}{\sqrt{2\omega_k}} \exp \left\{ -i \int^t \omega_k dt' \right\}$$

But again, since our initial conditions have $t = 0$ the above simplifies to just:

$$\chi_k(t_i) = \frac{1}{a(t_i)\sqrt{\omega_k}} = \frac{1}{\sqrt{\omega_k}}$$

Where we have used $a(t_i) = 1$. We can also differentiate χ_k modes to obtain initial conditions for $\dot{\chi}_k$, where we will only keep the adiabatic terms:

$$\dot{\chi} = [-H(t_i) - i\omega_k(t_i)]\chi_k(t_i) \approx 0$$

All of these initial conditions are dependent on parameters g, λ, f, v and technically t_i , though we set it to 0 previously. We can discuss the ranges of these parameters:

1. $10^{-5} < \lambda < 1$
2. $f = 5 \times 10^{-7}$
3. $10^{-14} < g < 1$
4. v is free

3 Integration strategy

First things first I need to change the variables from time to number of e-folds

$$t \rightarrow N, \quad \text{where} \quad N = \ln a$$

Then

$$\frac{dN}{dt} = \frac{\dot{a}}{a} = H \quad \Rightarrow \quad \frac{d\phi}{dt} = H \frac{d\phi}{dN} \quad \text{and} \quad \frac{d^2\phi}{dt^2} = H \frac{dH}{dN} \frac{d\phi}{dN} + H^2 \frac{d^2\phi}{dN^2}$$

In prime notation for any function A

$$\dot{A} = H A' \quad \text{and} \quad \ddot{A} = H^2 A'' + H' H A'$$

Then using Klein-Gordon, Friedman and acceleration equations

$$\begin{aligned} \ddot{\phi} + 3H\dot{\phi} + V_{,\phi} &= 0 \\ H^2 &= \frac{1}{3M_{\text{pl}}^2} \left(\frac{1}{2}\dot{\phi}^2 + V(\phi) \right) \\ \dot{H} &= -\frac{1}{2M_{\text{pl}}^2} \dot{\phi}^2 \end{aligned}$$

We rewrite the derivatives

$$\begin{aligned} H^2\phi'' + H'H\phi' + 3H^2\phi' + V_{,\phi} &= 0 \\ H^2 &= \frac{1}{3M_{\text{pl}}^2} \left(\frac{1}{2}H^2\phi'^2 + V(\phi) \right) \\ HH' &= -\frac{1}{2M_{\text{pl}}^2} H^2\phi' \end{aligned}$$

The last two equations can be rewritten in a more convenient form: in the first I'm just expressing H^2 in the second I'm expressing H'/H , which are the terms that will appear when we rearrange KG

$$H^2 = \frac{V(\phi)}{3M_{\text{pl}}^2 - \frac{1}{2}\phi'^2} \quad \text{and} \quad \frac{H'}{H} = -\frac{1}{2M_{\text{pl}}^2} \phi'^2$$

Then after dividing KG by H^2 and inserting these relations I obtain

$$\phi'' + \left(3 - \frac{\phi'^2}{2M_{\text{pl}}^2} \right) \phi' + \left(3M_{\text{pl}}^2 - \frac{1}{2}\phi'^2 \right) \frac{V_{,\phi}}{V} = 0$$

I'm not sure if I can do this, but I noticed that

$$W = \ln V \quad \Rightarrow \quad W_{,\phi} = \frac{V_{,\phi}}{V}$$

This is especially useful, since our potential is exponential, so using $V(\phi) = V_0 \exp \left[-\kappa e^{\frac{\phi}{bM_{\text{Pl}}}} \right]$ we get

$$W = -\kappa \exp\left(\frac{\phi}{bM_{\text{pl}}}\right) + W_0 \quad \text{and} \quad W_{,\phi} = -\frac{\kappa}{bM_{\text{pl}}} \exp\left(\frac{\phi}{bM_{\text{pl}}}\right)$$

where $W_0 = \ln V_0$. I can further massage this equation (Not sure if its worth to), by multiplying everything by ϕ' , then $\phi''\phi' = \frac{1}{2}(\phi'^2)'$ and $W_{,\phi}\phi' = bM_{\text{pl}}W'$, hence

$$\frac{1}{2}(\phi'^2)' + \left(3 - \frac{\phi'^2}{2M_{\text{pl}}^2}\right)\phi'^2 + \left(3M_{\text{pl}}^2 - \frac{1}{2}\phi'^2\right)bM_{\text{pl}}W' = 0$$

Taking $M_{\text{pl}} = 1$ and multiplying everything by 2

$$(\phi'^2)' + (6 - \phi'^2)\phi'^2 + (6 - \phi'^2)bW' = 0$$

Another approach, instead of multiplying everything by ϕ' we can just split these equations into a system with $u = \phi'$, then

$$\begin{cases} u = \phi' \\ u' + (6 - u^2)(u^2 + bW_{,\phi}) = 0 \end{cases}$$

Since I will be solving these equations numerically, I need some analytic results to check my calculations with. I will use slow-roll and kination approximations.

For slow-roll we consider that $|\dot{\phi}| \ll |3H\dot{\phi}|$ and $|\dot{\phi}| \ll |V_{,\phi}(\phi)|$, so Friedman equation simplifies to

$$H^2 = \frac{V(\phi)}{3M_{\text{pl}}^2}$$

And Slow-roll klein gordon (wrt time) equation becomes:

$$3H\dot{\phi} = -V_{,\phi}(\phi)$$

Again, using the same change of variables $t \rightarrow N$ we have

$$3H^2\phi' = -V_{,\phi} \quad \Rightarrow \quad \phi' = -M_{\text{pl}}^2 \frac{V_{,\phi}(\phi)}{V(\phi)} = -M_{\text{pl}}^2 W_{,\phi}$$

This now should be solvable analytically, again, setting $M_{\text{pl}} = 1$

$$N = \int -\frac{1}{W_{,\phi}} d\phi = \int \frac{b}{\kappa} \exp\left(-\frac{\phi}{b}\right) d\phi = -\frac{b^2}{\kappa} \exp\left(-\frac{\phi}{b}\right)$$

There is also the kination limit, which was discussed a bit earlier.